an eigenstate of an observable, then there is no uncertainty about the result of measuring that observable. But whatever the state, there is always uncertainty about some observable. If the state happens to be an eigenvector of one Hermitian operator—call it **A**—then it will not be an eigenvector of other operators that don't commute with **A**. Thus, as a rule, if **A** and **B** do not commute, then there must be uncertainty in one or the other, if not both.

The iconic example of this mutual uncertainty is the Heisenberg Uncertainty Principle, which in its original form had to do with the position and momentum of a particle. But Heisenberg's ideas can be expanded into a much more general principle that applies to any two observables that happen not to commute. An example would be two components of a spin. We now have all the ingredients necessary to derive the general form of the uncertainty principle.

5.4 THE MEANING OF UNCERTAINTY

We need to be very certain about what we mean by *uncertainty* if we want to quantify it. Let's suppose the eigenvalues of the observable **A** are called *a*. Then, given a state |Ψ>, there is a probability distribution *P* (*a*) with the usual properties. The expectation value of **A** is the ordinary average:

$$\langle \Psi | \mathbf{A} | \Psi \rangle = \sum_a a P(a).$$

Roughly speaking, this means that *P*(*a*) is centered around the expectation value. What we will mean by "the uncertainty in **A**" is the so-called *standard deviation*. To compute the standard deviation, begin by subtracting from **A** its expectation value. We define the operator $\bar{\mathbf{A}}$ to be:

$$\bar{\mathbf{A}} = \mathbf{A} - \langle \mathbf{A} \rangle.$$

By defining $\bar{\mathbf{A}}$ in this way, we have subtracted an expectation value from an operator, and it's not completely clear what that means. Let's take a closer